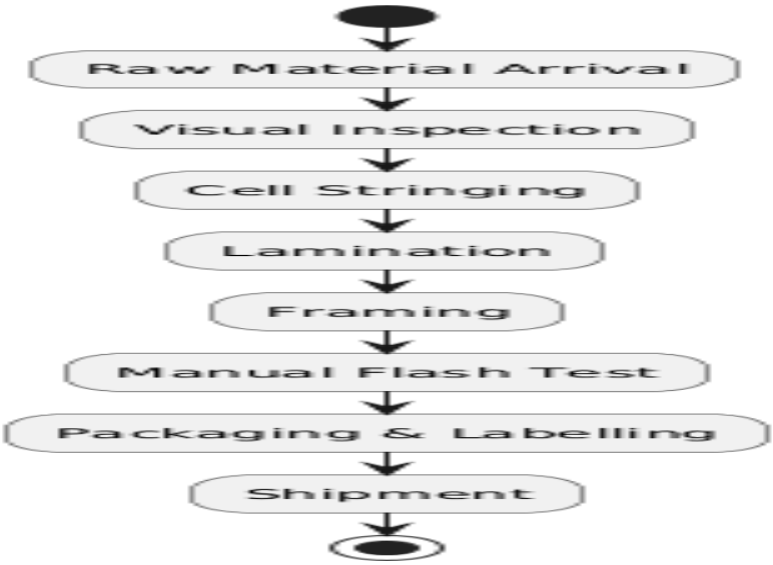
Process Flow Documentation

AS■IS / TO■BE Analysis

Prepared for: Solar Manufacturer | 02 May 2026

1. AS■IS Process (Current State)

Manual, paper■based tracking with spreadsheets. Operators record counts manually; quality tests are logged in notebooks or Excel. No real■time visibility, OEE is calculated after shift end, and traceability is batch■level at best.



2. TO■BE Process (Future State with MES)

Fully digital, AI■native. Sales orders flow into PODISE for real■time scheduling. Every machine data is captured via OPC UA and streamed through Kafka to the MES. Quality inspection is AI■powered (AOI/EL), and full genealogy is maintained. Dispatch is optimised with system■directed loading and ALMM COC auto■generation.

3. Key Differences

Aspect	AS■IS	TO■BE
Production tracking	Manual Excel/logs	Real■time SFDC + OEE dashboards
Quality	Paper forms, manual EL	AI■AOI/EL classification, flash test auto■capture
Scheduling	Planner’s spreadsheet	PODISE real■time APS with Gantt
Traceability	Batch level (no serial)	Full genealogy wafer→module; serial■level ALMM
Dispatch	Manual loading, paper COC	System■directed loading, digital COC